

Soumen Roy

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Experience

Centre for Cosmology, Particle Physics and Phenomenology (CP3), UCLouvain, Postdoctoral Researcher Louvain-la-Neuve, Belgium
2024 – present
2 years

- Active member of the LIGO–Virgo–KAGRA Collaboration.

Nikhef / Utrecht University, Postdoctoral Researcher The Netherlands
2020 – 2024
4 years

Education

Ph.D. Indian Institute of Technology Gandhinagar, Physics India
2014 – 2021

- Thesis: Aspects of gravitational wave data analysis from compact binary coalescences
- Supervisor: Prof. Anand S. Sengupta

M.Sc. The University of Calcutta, Physics (Astrophysics & Astroparticle Physics) India
2011 – 2013

- Thesis: An Introduction to Fluid Dynamics and Numerical Solution of Shock Tube Problem by Using Roe Solver
- Supervisor: Prof. Sanjay Kumar Ghosh

B.Sc. Vidyasagar University, Physics India
(Honours) 2008 – 2011

Computational Expertise

Programming and workflow: Python, C/C++, HTCondor, bash, LaTeX; experience with MATLAB, Mathematica, Gnuplot, and FORTRAN

Gravitational-wave software: LALSuite, PyCBC, Bilby, BayesWave; Bayesian inference and waveform-model implementation

Computing environments: Linux, macOS, Windows; high-throughput and high-performance computing for large-scale gravitational-wave analyses

Teaching

Spring 2017: Graduate Teaching Fellow, IIT Gandhinagar: Statistical Mechanics

Fall 2018: Graduate Teaching Fellow, IIT Gandhinagar: Computational Physics

Spring 2019: Graduate Teaching Fellow, IIT Gandhinagar: Introduction to Computing

Professional Memberships

Since 2020: Member of Einstein Telescope Observational Science Board

Since 2020: Member of Virgo Scientific Collaboration

Since 2017: Member of LIGO Scientific Collaboration

2017–2020: Member of LIGO-India Scientific Collaboration

Journal Referee

- Physical Review D, Physical Review Letters